

§ 1.4 Review

f is continuous $\Leftrightarrow \lim_{x \rightarrow a} f(x) = f(a) \Rightarrow \lim_{x \rightarrow a} (f(x) - f(a)) = 0$

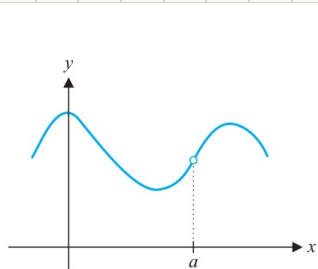


FIGURE 1.22a

$f(a)$ is not defined (the graph has a hole at $x = a$).

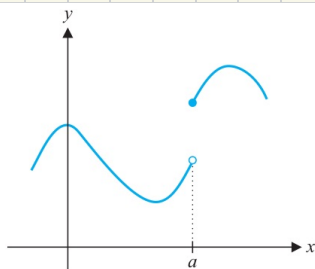


FIGURE 1.22b

$f(a)$ is defined, but $\lim_{x \rightarrow a} f(x)$ does not exist (the graph has a jump at $x = a$).

(jump discontinuous)

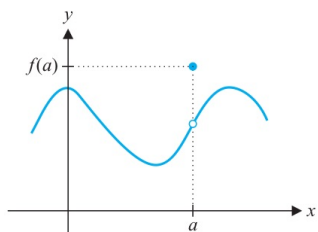


FIGURE 1.22c

$\lim_{x \rightarrow a} f(x)$ exists and $f(a)$ is defined, but $\lim_{x \rightarrow a} f(x) \neq f(a)$ (the graph has a hole at $x = a$).

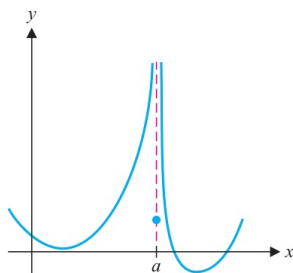


FIGURE 1.22d

$\lim_{x \rightarrow a} f(x)$ does not exist (the function "blows up" as x approaches a).

Continuous Functions on $\mathbb{R}$:

① Polynomials $x^5 + 4x^3 - 1$

② $\sin x$, $\cos x$

③ $\sqrt[n]{x}$ $\left\{ \begin{array}{l} n \text{ is odd: cont. on } \mathbb{R} \end{array} \right.$

$\parallel$
 $x^{\frac{1}{n}}$

$\left\{ \begin{array}{l} n \text{ is even: cont. on } x \geq 0. \end{array} \right.$

$$(x^{\frac{1}{3}})^3 = x$$

$$((-1)^{\frac{1}{2}})^2$$

THEOREM 4.2

Suppose that f and g are continuous at $x = a$. Then all of the following are true:

- (i) $(f \pm g)$ is continuous at $x = a$,
- (ii) $(f \cdot g)$ is continuous at $x = a$ and
- (iii) (f/g) is continuous at $x = a$ if $g(a) \neq 0$.

THEOREM 4.3

Suppose that $\lim_{x \rightarrow a} g(x) = L$ and f is continuous at L . Then,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(L).$$

COROLLARY 4.1

Suppose that g is continuous at a and f is continuous at $g(a)$. Then, the composition $f \circ g$ is continuous at a .

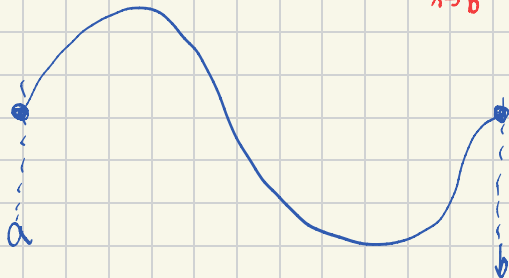
1. f is left cont. if $\lim_{x \rightarrow a^-} f(x) = f(a)$
2. f is right cont. if $\lim_{x \rightarrow a^+} f(x) = f(a)$
3. f is cont. if $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a) \iff \lim_{x \rightarrow a} f(x) = f(a)$

Def 4.2

1. f is continuous on $(a, b) \iff f$ is cont. at $x, \forall x \in (a, b)$
2. f is continuous on $[a, b] \iff f$ is cont. on (a, b)

$$\text{and } \lim_{x \rightarrow a^+} f(x) = f(a)$$

$$\text{and } \lim_{x \rightarrow b^-} f(x) = f(b)$$

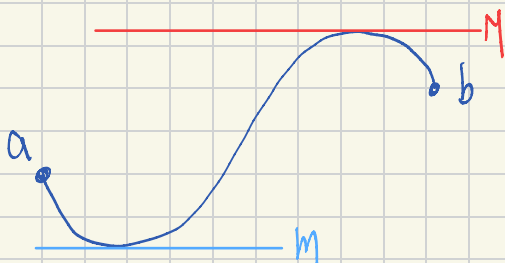


THEOREM 2.6.2 THE EXTREME-VALUE THEOREM

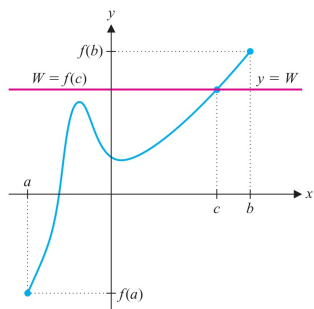
(Max-min Theorem)

(Salas Calculus)

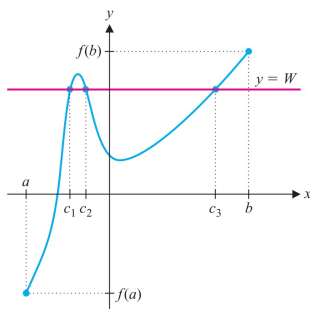
If f is continuous on a bounded closed interval $[a, b]$, then on that interval f takes on both a maximum value M and a minimum value m .

**THEOREM 4.4 (Intermediate Value Theorem)**

Suppose that f is continuous on the closed interval $[a, b]$ and W is any number between $f(a)$ and $f(b)$. Then, there is a number $c \in [a, b]$ for which $f(c) = W$.

**FIGURE 1.29a**

An illustration of the Intermediate Value Theorem

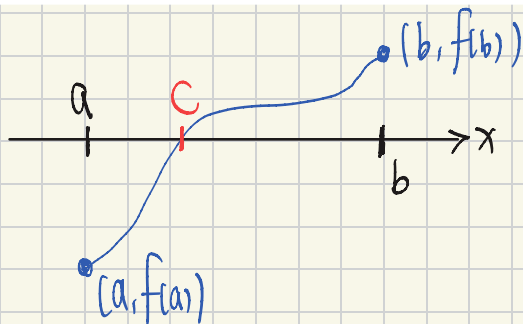
**FIGURE 1.29b**

More than one value of c

The Method of Bisections

COROLLARY 4.2

Suppose that f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs [i.e., $f(a) \cdot f(b) < 0$]. Then, there is at least one number $c \in (a, b)$ for which $f(c) = 0$. (Recall that c is then a zero of f .)



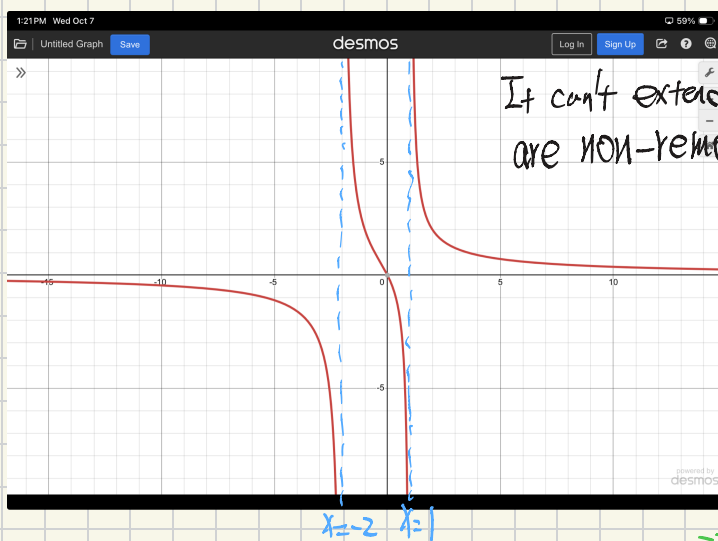
In exercises 1–14, determine where f is continuous. If possible, extend f as in example 4.2 to a new function that is continuous on a larger domain.

$$4. f(x) = \frac{4x}{x^2 + x - 2} = \frac{4x}{(x+2)(x-1)}$$

f is not defined if $(x+2)(x-1) = 0$

$$\Rightarrow x = -2 \text{ and } x = 1$$

Hence f is continuous on $(-\infty, -2) \cup (-2, 1) \cup (1, \infty)$
at $x \neq -2$ and $x \neq 1$.



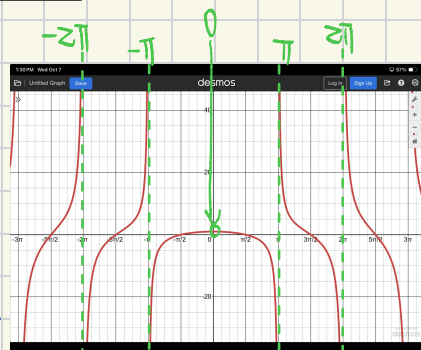
It can't extend since they are non-removable.

$$8. f(x) = x \cot x = \frac{x}{\tan x} = \frac{x \cos x}{\sin x}$$

f is not defined at $x = k\pi, k \in \mathbb{Z}$

$\Rightarrow f$ is not cont. at $x = k\pi, k \in \mathbb{Z}$

Since $x = k\pi, k \in \mathbb{Z}$, is non-removable, it can't extend.



$$1. \lim_{x \rightarrow c} f(x) \text{ exists}$$

$$2. f(c) \text{ is defined}$$

$$3. \lim_{x \rightarrow c} f(x) = f(c)$$

In exercises 15–20, explain why each function fails to be continuous at the given x -value by indicating which of the three conditions in Definition 4.1 are not met.

17. $f(x) = \sin \frac{1}{x}$ at $x=0$.

Since $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist, f is not cont. at $x=0$.

In exercises 21–26, determine the intervals on which f is continuous.

23. $f(x) = \frac{6}{\sqrt{x+1}}$

① $x+1 \geq 0$ ② $x+1 \neq 0$

① $x+1 \geq 0 \Rightarrow x \geq -1$

② $x+1 \neq 0 \Rightarrow x \neq -1$

Hence f is cont. when $x > -1$.
on $(-1, \infty)$.

27. Suppose that a state's income tax code states that the tax liability on x dollars of taxable income is given by

$$T(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 0.14x & \text{if } 0 < x < 10,000 \\ c + 0.21x & \text{if } 10,000 \leq x. \end{cases}$$

Determine the constant c that makes this function continuous for all x . Give a rationale why such a function should be continuous.

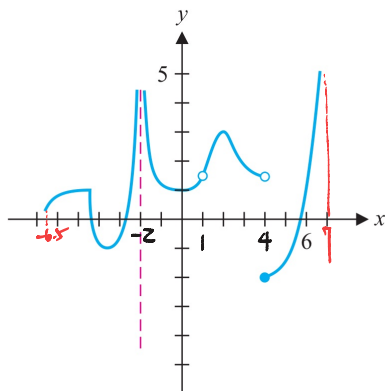
$$\lim_{x \rightarrow 10000^-} T(x) = \lim_{x \rightarrow 10000^-} 0.14x = 1400, \quad \lim_{x \rightarrow 10000^+} T(x) = \lim_{x \rightarrow 10000^+} c + 0.21x = c + 2100$$

$$\Rightarrow 1400 = c + 2100$$

$$\Rightarrow c = -700.$$

In exercises 35 and 36, use the given graph to identify all intervals on which the function is continuous.

35.



f is continuous on
 $[-6.5, -2) \cup [-2, 4) \cup (4, 6) \cup (6, 6.5)$

In exercises 37–39, determine values of a and b that make the given function continuous.

37. $f(x) = \begin{cases} \frac{2 \sin x}{x} & \text{if } x < 0 \\ a & \text{if } x = 0 \\ b \cos x & \text{if } x > 0 \end{cases}$ Consider $x = 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{2 \sin x}{x} = 2 \cdot 1 = 2, \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} b \cos x = b$$

① $2 = b$

② $2 = a$

$\Rightarrow \underline{a = b = 2}$

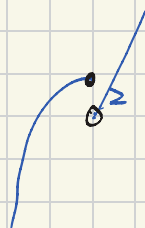
A function is continuous from the right at $x = a$ if $\lim_{x \rightarrow a^+} f(x) = f(a)$. In exercises 41 and 42, determine whether $f(x)$ is continuous from the right at $x = 2$.

41. $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ 3x - 3 & \text{if } x > 2 \end{cases}$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} 3x - 3 = 6 - 3 = 3$$

$$f(2) = 4$$

Since $\lim_{x \rightarrow 2^+} f(x) \neq f(2)$, f is not right cont. at $x = 2$.



44. Suppose that $f(x) = \frac{g(x)}{h(x)}$ and $h(a) = 0$. Determine whether each of the following statements is always true, always false or maybe true/maybe false. Explain. (a) $\lim_{x \rightarrow a} f(x)$ does not exist. (b) f is not continuous at $x = a$.

47. If f is continuous at $x = a$, prove that $g(x) = |f(x)|$ is continuous at $x = a$.

48. Determine whether the converse of exercise 47 is true. That is, if $|f|$ is continuous at $x = a$, is it necessarily true that f must be continuous at $x = a$?

50. If $f(x) = \begin{cases} x^2, & \text{if } x \neq 0 \\ 4, & \text{if } x = 0 \end{cases}$ and $g(x) = 2x$, show that

$$\lim_{x \rightarrow 0} f(g(x)) \neq f\left(\lim_{x \rightarrow 0} g(x)\right).$$

$$f(g(x)) = \begin{cases} 4x^2 & \text{if } x \neq 0 \\ 4 & \text{if } x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} f(g(x)) = 0$$

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} 2x = 0$$

$$f(0) = 4$$

$$\text{Since } 0 \neq 4, \lim_{x \rightarrow 0} f(g(x)) \neq f\left(\lim_{x \rightarrow 0} g(x)\right)$$

53. Prove that if f is continuous on an interval $[a, b]$, $f(a) > a$ and $f(b) < b$, then f has a fixed point (a solution of $f(x) = x$) in the interval (a, b) .